Overcoming the 'Impracticality' of RAG: Proposing a Real-World Benchmark and Multi-Dimensional Diagnostic Framework

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ABSTRACT

Performance evaluation of Retrieval-Augmented Generation (RAG) systems within enterprise environments is governed by multi-dimensional and composite factors extending far beyond simple final accuracy checks. These factors include reasoning complexity, retrieval difficulty, the diverse structure of documents, and stringent requirements for operational explainability. Existing academic benchmarks fail to systematically diagnose these interlocking challenges, resulting in a critical gap where models achieving high performance scores fail to meet the expected reliability in practical deployment. To bridge this discrepancy, this research proposes a multi-dimensional diagnostic framework by defining a four-axis difficulty taxonomy and integrating it into an enterprise RAG benchmark to diagnose potential system weaknesses.

Problems of Previous RAG Benchmarks

- **Lack of Diagnostic Capability:** Most existing benchmarks treat RAG systems as monolithic black boxes during evaluation, reporting only end-to-end QA accuracy as a measure of overall performance. This approach obscures the ability to systematically diagnose errors, making it difficult to discern whether a failure originates from the retriever's inability to fetch relevant context, or from the LLM's reasoning failure in generating a correct answer.
- **Lack of Evaluation of Composite Tasks:** Existing benchmarks often fail to give a comprehensive evaluation of RAG systems' capability of handling tasks requiring integration of multiple abilities, such as the ability to comprehending complex layouts, tables and charts, and collecting information scattered across a document.
- **Lack of Evidence Traceability:** Most existing bench-marks focus solely on evaluating the correctness of the final output, overlooking critical business requirements for the traceability and reliability of supporting evidence. Specifically, they lack evaluation dimensions that track the sources from which a RAG system generates its output and assess how faithfully the answer is grounded in those evidences.

A Survey of Previous RAG Benchmark Datasets

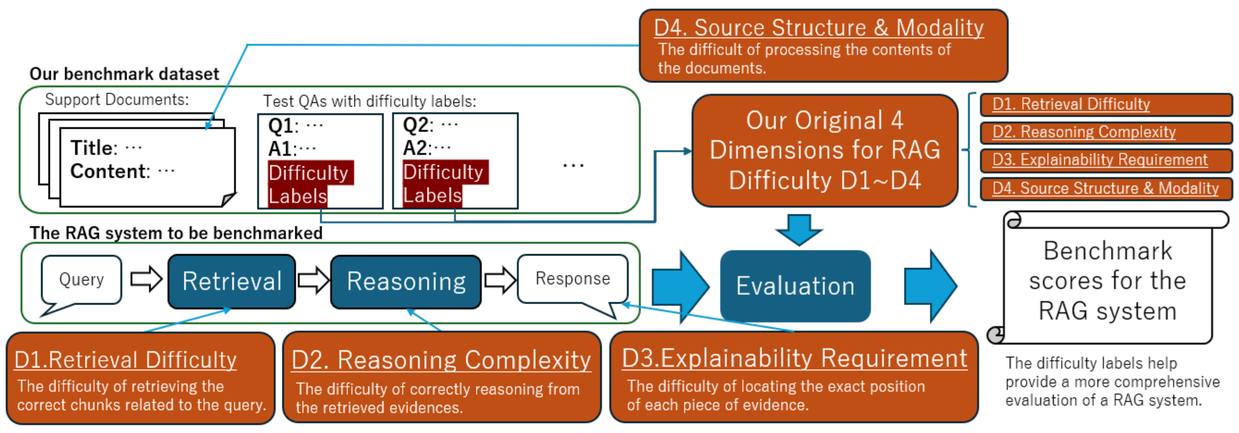
Dataset	Diagnostic Capability	Composite/Complex Tasks			Evidence Traceability
		Multi-doc	Multi-hop	Multimodal	
DomainRAG	Gen.	✓	48%	×	×
ETRQA	Gen.	✓	100%	×	×
HeteQA	Gen.	✓	18%	×	Text
JDocQA	Gen.	✓	15%	✓	Image
Multihop-RAG	Gen. & Retr.	✓	88%	×	Text
VisDoMBench	Gen. & Retr.	✓	100%	✓	Text & Image

- 3 main points required by Enterprise-RAG.
  - **Diagnostic Capability**
    - Can be achieved by introducing extra evaluation dimensions.
    - Previous methods mainly introduce two evaluations.
      - Generative Accuracy: End-to-end QA accuracy.
      - Retrieval Accuracy: How correctly the related chunks are retrieved.
  - **Composite/Complex Tasks**
    - Can be achieved by introducing tasks requiring complex or multiple abilities.
      - Multi-doc: Evidence scattered in multiple documents.
      - Multi-hop: Reasoning from multiple evidences required.
      - Multimodal: Non-plain-text (table, image, etc.) evidence included.
  - **Evidence Traceability**
    - Can be achieved by specifying the exact location of evidences.
      - Text Evidence: exact location.
      - Image Evidence: bounding box of the evidence.
- **None of these previous datasets** can cover all three points.
  - While VisDoMBench covers all types of difficult tasks, it does not annotate each test case with detailed difficulty tags, so it is still considered lack of diagnostic capability. (E.g., we may not pick up all multi-hop and multi-doc queries and evaluate the accuracy individually)

Our Proposed Benchmark Dataset

- Main idea:
  - **More diagnostic capability**: Annotate each query (test case) with detailed difficulty tags based on our own taxonomy of difficulty.
    - The taxonomy includes 22 difficulty tags classified into 4 dimensions. Details are listed in the chart at the top of the right column of this poster.
  - **More difficult**: More difficult task are introduced. E.g., retrieving from an image incorporating a complex layout like two-column page-formatting.
  - **Keep traceability**: Each query is annotated with detailed evidence location (bounding-box for image evidences) to ensure evidence traceability.

- Overview of the framework of our benchmark dataset.



- An example of a query in our dataset with its tags.

item	value
query	How do airborne communications and satellite communications differ in terms of A. how radio waves propagate, B. the amount of information that can be transmit-ted, and C. the difficulty of the technologies used?
answer	A. In aviation communications, radio waves can bend around obstacles, while satellite communications exhibit strong line-of-sight characteristics. B. Satel-lite communications can transmit larger amounts of information. C. Satellite communications involve more complex technology.
Bounding Box	top: 33.09, left: 48.84,width: 34.05,height: 25.95...
chunk in-fomation	Information and Communica- tions in Japan White Paper 2024 (English Version) :p.133

item	value
Reasoning Complexity	
Reasoning Depth	True(Multi-step Reasoning)
Quantitative Operation	False
Negation Question	False
Cause and Effect	False
Comparison (and Conditional Judgment)	True
Temporal Specification	False
Type of Output Processing	Summarization
Retrieval Difficulty	
multi-document	False
multi-chunk	False
Low Locality	False
Remote Reference	False
Document Volume	True(Over 1000 pages)
Chunk Size	False(512Token or less)
Abstraction Discrepancy	False
Vocabulary Mismatch	False
Source Structure & Modality	
Tables/Charts	True
Any Kind of Non-plain-text El-ements	True
Specific Area Reference	True
Logical Nesting	False
Large-Scale Enumeration	False
Redundancy	False
Explainability Requirement	
Strictness of Evidence Presenta-tion	Coordinate pre-sentation

Example of Benchmarking RAG Systems with our dataset

- Benchmarking with **end-to-end QA accuracy only**
  - Result: Agentic AI has better performance.
  - **Not clear:** Why it is better? **Which tasks** is it good at?

RAG System	JDocQA	Our Dataset
MRI-RAG	7.4%	9.0%
Agentic AI	14.9%	17.0%

- Benchmarking with our **original 4 dimensions of difficulty**.
  - Each score is computed by collecting all test cases classified as difficult with respect to the corresponding dimension (i.e., tagged with more than half of the tags of that dimension), and compute the QA accuracy of these difficult cases.

Dimension	MRI-RAG	Agentic AI
D1. Reasoning Complexity	2.5%	5.0% (+2.5%)
D2. Retrieval Difficulty	**0.0%**	12.0% **(+12.0%)**
D3. Source Structure & Modality	10.0%	20.0% **(+10.0%)**
D4. Explainability Requirement	9.1%	17.2% (+8.1%)

- **Now clear**: MRI-RAG has a weak ability in difficult retrieval tasks. It is also outperformed by Agentic AI in Source Structure & Modality by a large margin.
  - Developers can purposely improve their RAG system with this extra information.

Public Release of Our Dataset

Our dataset is planned to release on GitHub soon. Our first release will include 34 documents (1699 pages) and 100 queries. We sincerely invite those who are interested in this topic to collaborate with us to further enrich this dataset.